

JASON ZHI LIANG, Ph.D.

Principal Research Scientist | Automated AI Research | Self-Improving Agentic Systems

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Research Summary

Principal Research Scientist at Cognizant AI Lab, leading work on **automated AI research**, **open-ended learning**, and **self-improving agentic systems**, built on a decade of large-scale neuroevolution and evolutionary optimization for deep learning. My focus: automating the AI-research loop, designing architectures, tuning how models train, and now doing the research reasoning itself, with minimal human tuning, guided by two convictions: a system's founding values compound over its lifetime, and open-endedness and creativity must persist beyond learning, into ongoing use.

My work spans three layers of that stack: **CoDeepNEAT** (evolutionary neural architecture search, GECCO 2018; 1,500+ citations), **EPBT** (loss-function and hyperparameter evolution, GECCO 2021), and **Caesar** (Deep Research agent, ICML 2026 Human-AI Co-Creativity Workshop), one generate-and-test loop across all three: evolve and select architectures and training recipes, then critique and refine an agent's own reasoning. Caesar's adversarial refinement over a self-organizing knowledge graph outperforms the Gemini 3, Sonnet 4.5, and GPT-5.2 deep-research agents by 13–23% on a blinded, LLM-judged creative-synthesis benchmark. Named inventor on 5 issued U.S. patents in evolutionary AI and neural architecture search.

Technical Skills and Expertise

- **Research:** Inference-Time Compute Scaling, Deep Research Agents, Agentic Reasoning (System 2), Neural Architecture Search, Multi-Task Learning, Population-Based Training, Evolutionary Optimization for Deep Learning
- **Agentic Systems:** Multi-Agent Orchestration, Knowledge-Graph RAG and Reasoning, Stateful Agentic Memory, FSM-Based Coordination, LLM-as-a-Judge Evaluation, Evolutionary Prompt Optimization
- **Tooling:** PyTorch / TensorFlow, Frontier LLM APIs, Agentic Coding Tools (Claude Code), LlamaIndex, AutoGen / MetaGPT, Distributed GPU Clusters

Professional Experience

Cognizant AI Lab | Principal Research Scientist

San Francisco, CA | Dec 2018 – Present | Lead Researcher driving the roadmap for the lab's automated AI research effort.

1. Autonomous Self-Improvement Systems

- **Values-Based Self-Improvement Framework:** Autonomous self-improvement system built on three principles: (1) **values-first design**, starting values as the load-bearing axis for compounding self-improvement; (2) **safety-by-design**, an immutable safety floor the agent cannot rewrite; (3) **evaluator discipline**, every improvement claim verified by adversarial critique against frozen benchmarks before acceptance. This is early-stage research into how founding values shape long-horizon self-improvement.

2. Caesar: Autonomous Reasoning & Knowledge Synthesis

- **Automated R&D via Inference-Time Self-Improvement:** Designed and implemented “Caesar,” an autonomous Deep Research agent. Scaled test-time compute using an adversarial refinement loop that actively critiques internal drafts, generates orthogonal queries to target weaknesses, and consolidates findings via a generative merge. **Outperforms the Gemini 3, Sonnet 4.5, and GPT-5.2 deep-research agents by 13–23% across output formats on a blinded, LLM-judged creative-synthesis benchmark (Cliff’s $\delta \geq 0.76$).**
- **Graph-Based Associative Reasoning:** Replaced static RAG retrieval with a dynamic “Perceive-Think-Act” loop backed by a self-organizing knowledge graph, enabling discovery of non-obvious cross-disciplinary connections beyond the reach of linear retrieval agents.
- **Open-Ended Exploration Policy:** Designed an active information-foraging policy that detects stagnation in memory and graph structures, autonomously executing strategic backtracking to maximize information gain over long horizons. **Ablation: reducing exploration budget from 1000 to 250 iterations causes up to -3.84 total-score decline on long-form answers, validating that deep topological traversal is essential for creative synthesis.**

3. Multi-Agent Systems & Large-Scale Evolutionary Optimization

- **Multi-Agent Code Generation:** Developed a **Finite State Machine (FSM)** framework to coordinate teams of specialized coding agents. Applied evolutionary algorithms to optimize the *composition* of these teams, evaluated on SciCode and HumanEval+ code-generation benchmarks.
- **Evolutionary Population-Based Training (EPBT):** Pioneered a population-based approach that evolves loss functions and hyperparameters jointly with weights via population-based weight sharing. **Outperforms the PBT baseline by 1.3pp on CIFAR-10 ResNet-32 (92.79% vs 91.53%) while exploring 520 candidate loss functions at $12.5\times$ lower search cost than independent training (GECCO 2021; U.S. patent pending).**
- **Code Repository Analysis:** Created “Code Archaeologist,” an agentic system capable of reasoning over repository histories (via Git/LFS) to detect architectural patterns and technical debt.

Sentient Technologies | Research Scientist (prev. Research Intern)

San Francisco, CA | Dec 2015 – Nov 2018

- **Massive-Scale Evolutionary Architecture Search:** Scaled **CoDeepNEAT** (evolutionary neural architecture search, developed during my UT Austin PhD) to clusters of hundreds of GPUs. Validated that population-based search scales with parallel compute; established

asynchronous evolutionary optimization at scale and forms the basis for multiple issued U.S. patents.

- **Multi-Task Learning:** Achieved state-of-the-art on the **20-task Omniglot multi-task learning benchmark (88% accuracy, +21pp over Soft Ordering at 67%)** by evolving modular topologies with cross-task weight sharing.
- **Production Deployment:** Collaborated with Sentient’s product team to productionize CoDeepNEAT for image captioning.

The University of Texas at Austin | Research Assistant / Ph.D. Candidate

Austin, TX | Sep 2013 – Dec 2018

- **Solving Sparse Reward Problems:** Developed **MEA (Meta-Evolutionary Algorithm)**, a bilevel optimization framework. Successfully applied this to complex continuous control tasks (e.g., helicopter hovering), solving non-differentiable control problems where standard Gradient-based RL struggled to converge.

Selected Publications & Patents

Full publication list available on Google Scholar

Caesar: Deep Agentic Web Exploration for Creative Answer Synthesis (2026) *Jason Liang, Elliot Meyerson, Risto Miikkulainen*. [arXiv:2604.20855](https://arxiv.org/abs/2604.20855). Accepted at the **Human-AI Co-Creativity Workshop at ICML 2026**.

- Autonomous R&D agent that scales inference-time compute through an adversarial refinement loop over a self-organizing knowledge graph, outperforming leading Deep Research products on a blinded, LLM-judged creative-synthesis benchmark.

Asynchronous Evolution of Deep Neural Network Architectures (2024) *Jason Liang, Hormoz Shahrzad, Risto Miikkulainen*, *Applied Soft Computing*, Vol. 152.

- Techniques for asynchronous population-based optimization at scale, essential for efficient training on massive distributed clusters.

Regularized Evolutionary Population-Based Training (2021) *Jason Liang, Santiago Gonzalez, Hormoz Shahrzad, and Risto Miikkulainen*, *GECCO 2021*.

- Method to dynamically optimize loss functions during training, improving convergence speed and generalization.

Evolutionary Neural AutoML for Deep Learning (2019) *Jason Liang, Elliot Meyerson, Babak Hodjat, et al.*, *GECCO 2019*.

- Evolutionary optimization of novel neural network architectures for multi-task learning.

Evolutionary Architecture Search for Deep Multitask Networks (2018) *Jason Liang, Elliot Meyerson, and Risto Miikkulainen*, *GECCO 2018*.

- Novel evolutionary algorithm (CoDeepNEAT) that automates the discovery of deep neural architectures, outperforming human-designed baselines.

Evolving Deep Neural Networks (2019) *Risto Miikkulainen, Jason Liang, Elliot Meyerson, et al.*, Artificial Intelligence in the Age of Neural Networks and Brain Computing.

- Book chapter surveying neuroevolutionary methods for deep learning; introduces CoDeepNEAT and demonstrates applications to language modeling, image captioning, and multi-task learning.

Evolutionary Bilevel Optimization for Complex Control Tasks (2015) *Jason Zhi Liang, Risto Miikkulainen*, GECCO 2015.

- Solving complex control tasks using evolutionary bilevel optimization, handling sparse reward signals where traditional RL failed.

Patents (Issued, U.S.):

- **US Patent 12,282,845:** Multi-objective Coevolution of Deep Neural Network Architectures (2025).
- **US Patent 11,507,844:** Asynchronous Evaluation Strategy for Evolution of Deep Neural Networks (2022).
- **US Patent 11,250,328:** Cooperative Evolution of Deep Neural Network Structures (2022).
- **US Patent 11,030,529:** Evolution of Architectures for Multitask Neural Networks (2021).
- **US Patent 11,003,994:** Evolutionary Architectures for Evolution of Deep Neural Networks (2021).

Patents (Pending, U.S.):

- **App. 17/389,961:** Regularized Evolutionary Population-Based Training (2023).
- **App. 17/064,706:** Sharing Meta-Learning Methods Among Multiple Private Data Sets (2022).

Awards & Honors

- **BEACON Research Grant** (2015), BEACON Center for the Study of Evolution in Action (an NSF Science & Technology Center).

Professional Activities

- **ICML Silver Reviewer**, International Conference on Machine Learning (ICML) | 2026
- **Reviewer**, Genetic and Evolutionary Computation Conference (GECCO) | 2019 – Present

Education

Ph.D. in Computer Science · *Neuroevolution & Deep Learning*

The University of Texas at Austin · Sep 2013 – Dec 2018 · Advisor: Risto Miikkulainen

Dissertation: *Evolutionary Neural Architecture Search for Deep Learning*

B.S. in Electrical Engineering and Computer Science

University of California, Berkeley · Sep 2009 – May 2013